

# fixed

## Fixed

**Revision:** 220 **Last modified:** 2026-06-16T20:36:13Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db\_to\_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: **803**.

| HXC      | Type    | Pri | Title                                                                     |
|----------|---------|-----|---------------------------------------------------------------------------|
| HXC-001  | Task    | P1  |                                                                           |
| HXC-002  | Task    | P1  |                                                                           |
| HXC-003  | Task    | P1  |                                                                           |
| HXC-004  | Task    | P1  |                                                                           |
| HXC-005  | Task    | P1  |                                                                           |
| HXC-006  | Task    | P1  |                                                                           |
| HXC-007  | Task    | P1  |                                                                           |
| HXC-008  | Task    | P1  |                                                                           |
| HXC-009  | Task    | P1  |                                                                           |
| HXC-010  | Task    | P1  |                                                                           |
| HXC-011  | Task    | P1  |                                                                           |
| HXC-012  | Task    | P1  |                                                                           |
| HXC-013  | Task    | P1  |                                                                           |
| HXC-014  | Task    | P1  |                                                                           |
| HXC-015  | Task    | P1  |                                                                           |
| HXC-1000 | Bug     | P0  | FIX delta-ORSet strong-eventual-consistency defect (remove-before-add)    |
| HXC-1001 | Task    | P0  | Stand up embedded-etcd integration harness for pkg/etcd tests             |
| HXC-1002 | Feature | P0  | Prove etcd Put/Get round-trip and missing-key semantics against real etcd |
| HXC-1003 | Feature | P1  | Prove etcd GetPrefix returns exactly matching multi-key set               |
| HXC-1004 | Feature | P1  | Prove etcd Delete/DeletePrefix remove keys (sink-side absence)            |
| HXC-1005 | Feature | P0  | Prove etcd Watch fires correct event and closes on ctx cancel             |
| HXC-1006 | Feature | P1  | Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior          |
| HXC-1007 | Feature | P0  | Prove etcd distributed Lock mutual exclusion (highest-risk path)          |
| HXC-1008 | Task    | P2  | Add mutation-paired etcd New() default-timeout injection test             |

| HXC      | Type    | Pri | Title                                                                                    |
|----------|---------|-----|------------------------------------------------------------------------------------------|
| HXC-1009 | Task    | P2  | Prove etcd error-wrapping on dead endpoint                                               |
| HXC-1010 | Feature | P0  | Replace pkg/infra map-echo orchestrator with real service Boot via tests                 |
| HXC-1011 | Feature | P0  | Make pkg/infra Health probe real services and report false on failure                    |
| HXC-1012 | Feature | P1  | Make pkg/infra Logs capture real sink-side output honoring tail/since/timestamp          |
| HXC-1013 | Feature | P1  | Make pkg/infra Scale create real replicas verified via runtime query                     |
| HXC-1014 | Feature | P1  | Make pkg/infra VMSpawn/VMSSH establish a real session end-to-end                         |
| HXC-1015 | Feature | P2  | Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observability                |
| HXC-1016 | Feature | P0  | Add EtcdLocker integration tests against real etcd (block/acquire/lease)                 |
| HXC-1017 | Bug     | P0  | Rewrite TestMemoryLockerConcurrent to detect a broken lock without a lock                |
| HXC-1018 | Task    | P2  | Add explicit block-then-acquire ordering test for MemoryLocker                           |
| HXC-1019 | Bug     | P2  | Characterize or fix MemoryLocker.Lock context-cancellation goroutine                     |
| HXC-1020 | Feature | P0  | Wire a real Builder behind pkg/build Service and prove image is pullable                 |
| HXC-1021 | Bug     | P1  | Drive pkg/build StateFailed via a genuine build failure not the 'fail' sentinel          |
| HXC-1022 | Bug     | P1  | Make pkg/build List/Concurrent tests Start() the service and assert termination          |
| HXC-1023 | Task    | P2  | Replace time.Sleep polling in pkg/build with completion sync and add some mutation tests |
| HXC-1024 | Bug     | P2  | Enforce or document pkg/build content-addressable cache digest integrity                 |
| HXC-1025 | Task    | P2  | Cover pkg/build cancel-mid-flight branch sets StateCancelled                             |
| HXC-1026 | Feature | P0  | Add real mTLS handshake test for pkg/security (valid client succeeds, non-rejected)      |
| HXC-1027 | Bug     | P1  | Assert MinVersion==TLS13 on the client TLS path in pkg/security                          |
| HXC-1028 | Feature | P0  | Locate/build production Vault client and add dev-Vault integration test (CI)             |
| HXC-1029 | Task    | P3  | Replace pkg/security non-nil-only TestNewTLSConfigBuilder with meaningful                |
| HXC-1030 | Bug     | P1  | Verify pkg/hxcregistry item_history audit-sink rows after Create/Update/Delete           |
| HXC-1031 | Bug     | P2  | Assert exact and stable ComputeHeadingHash value in pkg/hxcregistry                      |
| HXC-1032 | Task    | P2  | Add pkg/hxcregistry failure-path tests (dup PK, missing-id Get/Update)                   |
| HXC-1033 | Bug     | P2  | Make pkg/hxcregistry concurrent test do concurrent writes (or rename)                    |
| HXC-1034 | Task    | P3  | Prove pkg/hxcregistry timestamp round-trip and DB CHECK-constraint                       |
| HXC-1035 | Bug     | P0  | Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove it                        |
| HXC-1036 | Task    | P1  | Add pkg/wasm negative tests for missing execute export and corrupt module                |
| HXC-1037 | Bug     | P2  | Prove pkg/wasm init/shutdown exports actually run via memory side-effects                |
| HXC-1038 | Task    | P3  | Cover pkg/wasm Host.Call ErrUnexpectedResult branch                                      |
| HXC-1039 | Bug     | P1  | Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany                        |
| HXC-1040 | Feature | P1  | Add pkg/semaphore M>>N concurrency invariant test under -race                            |
| HXC-1041 | Task    | P3  | Add pkg/semaphore New(0) barrier and exact-capacity boundary tests                       |
| HXC-1042 | Bug     | P1  | Convert pkg/storage error tests to errors.Is on exported sentinels                       |
| HXC-1043 | Bug     | P2  | Reconcile and test pkg/storage FileStore vs MemoryStore copy semantics                   |
| HXC-1044 | Task    | P2  | Add pkg/storage overwrite, missing-delete, and List edge tests for both s                |
| HXC-1045 | Feature | P1  | Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex                    |
| HXC-1046 | Bug     | P1  | Gate -race in CI for pkg/metrics concurrency mutation tests                              |

| HXC      | Type    | Pri | Title                                                                            |
|----------|---------|-----|----------------------------------------------------------------------------------|
| HXC-1047 | Bug     | P2  | Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, or            |
| HXC-1048 | Task    | P2  | Add pkg/metrics duplicate-registration and histogram edge tests                  |
| HXC-1049 | Task    | P3  | Add pkg/metrics /metrics integration test via httptest.Server                    |
| HXC-1050 | Bug     | P1  | Test pkg/grpcutil LoggingStreamInterceptor recv/send counters                    |
| HXC-1051 | Bug     | P2  | Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fi          |
| HXC-1052 | Task    | P2  | Cover pkg/grpcutil status.FromError code extraction for handler errors           |
| HXC-1053 | Bug     | P1  | Add pkg/context TestDetach_PreservesValues for the value-retention gu            |
| HXC-1054 | Bug     | P2  | Strengthen pkg/context WithTimeout deadline and expiry assertions                |
| HXC-1055 | Bug     | P1  | Give pkg/workerpool TestPool_NilWorkSafe_Mutation a real sink                    |
| HXC-1056 | Feature | P1  | Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recover, tests       |
| HXC-1057 | Bug     | P1  | Make pkg/errors concurrency test honest via locked GetFields path + -ra          |
| HXC-1058 | Task    | P2  | Pin pkg/errors Code enum exact values, distinctness, and WithFields nil          |
| HXC-1059 | Bug     | P2  | Replace tautological pkg/backoff cap-mutation test and pin Default()             |
| HXC-1060 | Bug     | P2  | Assert pkg/classads parser AST Op per operator and paren precedence              |
| HXC-1061 | Task    | P3  | Add pkg/classads eval failure-path and regexp non-match tests                    |
| HXC-1062 | Bug     | P2  | Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is c             |
| HXC-1063 | Task    | P3  | Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBI             |
| HXC-1064 | Bug     | P1  | Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug             |
| HXC-1065 | Task    | P3  | Strengthen pkg/log Warn test, add Debug-suppression, remove dead cap scaffolding |
| HXC-1066 | Task    | P3  | Add pkg/lru boundary, cold-miss, and struct-value round-trip tests               |
| HXC-1067 | Feature | P2  | Add pkg/pubsub -race concurrency test and explicit drop-boundary asse            |
| HXC-1068 | Bug     | P2  | Prove pkg/ratelimit bucket starts full and add Limiter conformance + -ra         |
| HXC-1069 | Bug     | P1  | Bring pkg/retry DoWithResult to behavioral parity with Do                        |
| HXC-1070 | Task    | P2  | Add pkg/retry transient-recovery, jitter-bound, and computeDelay table           |
| HXC-1071 | Task    | P3  | Strengthen pkg/serde MustMarshal exact-content and add Unmarshal fa              |
| HXC-1072 | Bug     | P2  | Pin pkg/validator error contract via errors.Is and remove/wire unused se         |
| HXC-1073 | Task    | P3  | Add pkg/validator uint min/max, IsValidID empty, oneof/email boundar             |
| HXC-1074 | Task    | P0  | Establish fleet-wide -race CI gate across concurrency-sensitive packages         |
| HXC-1075 | Feature | P0  | Implement NATS/JetStream client wrapper (pkg/events)                             |
| HXC-1076 | Feature | P0  | Implement etcd client wrapper (pkg/etcd)                                         |
| HXC-1077 | Feature | P0  | Implement etcd discovery backend (pkg/discovery)                                 |
| HXC-1078 | Feature | P0  | Implement real Kafka producer/consumer (pkg/pubsub)                              |
| HXC-1079 | Feature | P0  | Implement cgroups v2 + /proc resource reader (pkg/resources)                     |
| HXC-1080 | Feature | P0  | Implement Omega-model scheduler core (pkg/scheduler, internal/schedu             |
| HXC-1081 | Feature | P0  | Implement optimistic-concurrency versioning for scheduler pool                   |
| HXC-1082 | Feature | P1  | Implement HTCondor ClassAds matching in scheduler (pkg/classads)                 |
| HXC-1083 | Feature | P1  | Implement gang scheduling and preemption in scheduler                            |
| HXC-1084 | Feature | P0  | Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/              |

| HXC      | Type    | Pri | Title                                                                              |
|----------|---------|-----|------------------------------------------------------------------------------------|
| HXC-1085 | Feature | P0  | Implement CRDT session state and lifecycle (create/attach/migrate)                 |
| HXC-1086 | Feature | P0  | Implement real Node gRPC handlers serving helixv1 Node API (internal/node)         |
| HXC-1087 | Feature | P0  | Make node registry etcd-backed and cross-process shared (internal/node)            |
| HXC-1088 | Feature | P0  | Implement SWIM gossip membership and failure detection (pkg/swim)                  |
| HXC-1089 | Feature | P0  | Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)               |
| HXC-1090 | Feature | P1  | Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)                |
| HXC-1091 | Feature | P0  | Wire API gateway reverse proxy with prefix routing and /health (internal/gateway)  |
| HXC-1092 | Feature | P0  | Add JWT/RBAC auth middleware to API gateway (internal/gateway)                     |
| HXC-1093 | Feature | P0  | Implement leader election with fencing tokens (pkg/leader)                         |
| HXC-1094 | Feature | P0  | Implement distributed lock primitives over etcd (pkg/lock)                         |
| HXC-1095 | Feature | P1  | Implement Postgres-backed internal artifact registry (pkg/hxcregistry)             |
| HXC-1096 | Feature | P1  | Implement GPU resource probe with /proc parser (internal/gpu)                      |
| HXC-1097 | Feature | P1  | Define vendor-agnostic GPUBackend interface and registry (internal/gpu)            |
| HXC-1098 | Feature | P2  | Implement GPU sharing modes (MPS, time-slice, MIG, exclusive)                      |
| HXC-1099 | Feature | P1  | Implement Build Service orchestrator and worker pool (internal/build)              |
| HXC-1101 | Feature | P0  | Implement e2ee transport record protocol (security/pkg/e2ee)                       |
| HXC-1102 | Feature | P1  | Implement software-rooted attestation (security/pkg/attestation)                   |
| HXC-1103 | Feature | P0  | Implement RBAC scopes and identity bindings (internal/security)                    |
| HXC-1104 | Feature | P1  | Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)         |
| HXC-1105 | Feature | P0  | Implement htmux CLI raw-mode terminal client (cmd/htmux)                           |
| HXC-1106 | Feature | P0  | Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)            |
| HXC-1107 | Feature | P0  | Implement WebSocket envelope and MessagePack framing (pkg/websocket)               |
| HXC-1108 | Feature | P1  | Implement HelixQA live-service Challenge runner (cmd/helix-test)                   |
| HXC-1109 | Feature | P1  | Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test)   |
| HXC-1110 | Task    | P1  | Wire Prometheus /metrics endpoint uniformly across all services (cmd/helix-test)   |
| HXC-1111 | Task    | P1  | Wire Jaeger/OTel trace-ID propagation across services (pkg/tracing)                |
| HXC-1112 | Task    | P1  | Wire pkg/health gRPC health protocol on every service (pkg/health, cmd/helix-test) |
| HXC-1113 | Bug     | P2  | Remove dangling pkg/gpuattest placeholder and reuse security/pkg/gpuattest         |
| HXC-1114 | Feature | P0  | Provision PostgreSQL primary schema with 15 tables, indexes, and triggers          |
| HXC-1115 | Task    | P0  | Define etcd key namespace and key-builder constants (pkg/etcd)                     |
| HXC-1116 | Feature | P1  | Define Redis cache, routing, and rate-limit key structure (pkg/storage)            |
| HXC-1117 | Task    | P0  | Define and generate proto service stubs (node/session/scheduler/health/alerts)     |
| HXC-1118 | Feature | P1  | Implement REST API surface on API Gateway with OpenAPI 3.0                         |
| HXC-1119 | Feature | P1  | Implement Avro event schemas and schema-validated routing (Event Bus)              |
| HXC-1120 | Task    | P1  | Configure Kafka topics with partitions/replication/retention                       |
| HXC-1121 | Task    | P0  | Configure NATS JetStream streams (HELIX_NODES/SESSIONS/SCHEDULING/HEALTH/ALERTS)   |
| HXC-1122 | Feature | P2  | Implement Cap'n Proto zero-copy serialization for control-plane (pkg/session)      |
| HXC-1123 | Feature | P2  | Implement FlatBuffers serialization for GPU compute payloads                       |
| HXC-1124 | Feature | P2  | Implement ZeroMQ message patterns for internal data planes                         |

| HXC      | Type     | Pri | Title                                                                        |
|----------|----------|-----|------------------------------------------------------------------------------|
| HXC-1125 | Feature  | P1  | Implement Setup Wizard single-command node onboarding (cmd, BAS)             |
| HXC-1126 | Feature  | P2  | Implement LLM Brain advisory engine with LLMsVerifier gate (internal)        |
| HXC-1127 | Feature  | P1  | Implement OPA/WASM Policy Engine with HelixConstitution enforcement          |
| HXC-1130 | Feature  | P2  | Implement Zellij and GNU screen session backends                             |
| HXC-1131 | Feature  | P2  | Implement Backup Service (etcd snapshots, WAL archival, Ceph checkpoints)    |
| HXC-1132 | Feature  | P1  | Implement Metrics Collector node scrape endpoint and GPU metrics aggregation |
| HXC-1133 | Feature  | P1  | Implement token-bucket rate limiter for API gateway (pkg/ratelimit)          |
| HXC-1134 | Feature  | P2  | Implement Vault secret injection and rotation for all services               |
| HXC-1135 | Feature  | P0  | Implement two-node cluster formation end-to-end (exit gate)                  |
| HXC-1136 | Feature  | P0  | Implement session-create end-to-end path (htmux new exit gate)               |
| HXC-1137 | Task     | P1  | Benchmark job scheduling decision latency under 100ms (exit gate)            |
| HXC-1138 | Task     | P1  | Enforce >60% pkg/ line coverage and paired mutation tests (Constitution)     |
| HXC-1139 | Research | P2  | Author TLA+ specifications for consensus and scheduling concurrency          |
| HXC-1141 | Feature  | P2  | Implement interactive-mode AI CLI agent resource provisioning                |
| HXC-1142 | Feature  | P2  | Implement hybrid mode switching and batch/interactive resource sharing       |
| HXC-1144 | Feature  | P2  | Implement WireGuard mesh network policies (segmentation)                     |
| HXC-1145 | Docs     | P2  | Author MVP architecture and component-specification documentation and        |
| HXC-1146 | Docs     | P0  | Document node-provisioning boundary: project does NOT implement jail         |
| HXC-1147 | Feature  | P1  | Implement internal/console/linux_boot.go BootCoordinator state machine       |
| HXC-1148 | Feature  | P1  | Replace pkg/wireguard NAT discovery with RFC 5389 STUN client                |
| HXC-1149 | Task     | P3  | Return typed ErrUnsupported for UPnP/NAT-PMP in pkg/wireguard                |
| HXC-1150 | Feature  | P1  | Add WireGuard key-rotation grace/overlap window with ActiveKeys()            |
| HXC-1151 | Task     | P1  | Add pkg/discovery etcd-backed integration test against a real etcd           |
| HXC-1152 | Feature  | P1  | Implement pkg/session container-checkpoint migration strategy (CRDT)         |
| HXC-1156 | Feature  | P1  | Implement thermal-aware scheduler plugin (throttle on overheat)              |
| HXC-1157 | Feature  | P1  | Implement semi-trusted node output verification (redundant-compute/ch        |
| HXC-1158 | Task     | P2  | Harden device hardware-identification in internal/console/detector.go (g     |
| HXC-1160 | Feature  | P2  | Improve tmux session backend to full PTY wrap on Attach                      |
| HXC-1162 | Task     | P2  | E2E: multi-node session CRDT convergence                                     |
| HXC-1164 | Docs     | P2  | Author phase_02 architecture diagram (operator console + mesh + node         |
| HXC-1165 | Task     | P2  | Ensure api/v1/node.proto carries trust_level + thermal fields consumed e     |
| HXC-1167 | Task     | P0  | Build ARM64 cross-compilation toolchain for Helix agent                      |
| HXC-1175 | Feature  | P0  | Implement charging-gated PowerGater scheduling guard                         |
| HXC-1178 | Feature  | P0  | Implement MQTT client for edge work dispatch and status                      |
| HXC-1184 | Feature  | P0  | Implement edge node registration for all tiers (T3-T8)                       |
| HXC-1185 | Feature  | P0  | Implement edge heartbeat with battery/thermal/network telemetry              |
| HXC-1186 | Feature  | P0  | Implement edge protocol gateway (MQTT/QUIC/WebSocket)                        |
| HXC-1187 | Feature  | P0  | Define EdgeWorkUnit and EdgeWorkResult protobuf schemas                      |

| HXC      | Type     | Pri | Title                                                                                                      |
|----------|----------|-----|------------------------------------------------------------------------------------------------------------|
| HXC-1188 | Feature  | P0  | Enforce work-unit resource limits (duration/memory/CPU) on edge device                                     |
| HXC-1189 | Feature  | P0  | Implement EdgeAwarePlugin scheduler Filter stage                                                           |
| HXC-1190 | Feature  | P1  | Implement EdgeAwarePlugin scheduler Score stage                                                            |
| HXC-1191 | Feature  | P1  | Implement declarative per-tier ScheduleRule engine                                                         |
| HXC-1193 | Feature  | P1  | Implement per-tier workload quantization selection                                                         |
| HXC-1195 | Feature  | P0  | Implement edge trust-level model (STANDARD/SEMI/EDGE_DONOR)                                                |
| HXC-1196 | Feature  | P0  | Enforce workload restriction matrix by trust level                                                         |
| HXC-1197 | Feature  | P0  | Implement edge output verification (LLMsVerifier/redundant/checksum)                                       |
| HXC-1198 | Feature  | P1  | Implement offline sync protocol with delta compression                                                     |
| HXC-1199 | Feature  | P1  | Implement edge sensor-fusion framework and stream workload type                                            |
| HXC-1201 | Feature  | P2  | Implement low-bandwidth protocol optimizations (CoAP option)                                               |
| HXC-1202 | Research | P3  | Implement WebRTC P2P data-channel transport for edge devices                                               |
| HXC-1207 | Task     | P0  | Implement edge device chaos test suite                                                                     |
| HXC-1210 | Bug      | P0  | Wire EtcdBackend into the node agent for cluster-wide discovery                                            |
| HXC-1211 | Bug      | P0  | Make internal/scheduler.StreamJobEvents a real event stream                                                |
| HXC-1212 | Feature  | P0  | Connect pkg/classads to scheduler Filter/Score stages                                                      |
| HXC-1213 | Feature  | P0  | Add auth + RBAC enforcement to the gateway                                                                 |
| HXC-1214 | Feature  | P1  | Add S3-compatible (minio) backend to pkg/storage                                                           |
| HXC-1215 | Feature  | P1  | Add OTLP/Jaeger export shim for pkg/tracing                                                                |
| HXC-1216 | Feature  | P2  | Implement true embedded Raft consensus (pkg/raft)                                                          |
| HXC-1220 | Task     | P2  | Replace custom policy engine with OPA/Rego                                                                 |
| HXC-1229 | Task     | P0  | Build device profile registry with versioned YAML schema for T1-T8                                         |
| HXC-1230 | Feature  | P1  | Implement automated tier detection validating host KVM/ARM64/binfmt                                        |
| HXC-1231 | Feature  | P1  | Implement device provisioning lifecycle abstraction (Provisioner/Instantiator)                             |
| HXC-1233 | Feature  | P1  | Implement qcow2 copy-on-write overlay management with 10-deep chain                                        |
| HXC-1236 | Feature  | P0  | Implement INetwork/HelixNetwork trait with production and simulation                                       |
| HXC-1237 | Feature  | P0  | Wire DST sim transport seam into pkg/swim membership (prod+sim parallel)                                   |
| HXC-1238 | Feature  | P1  | Implement BUGGIFY macro framework with ~25% deterministic fire rate                                        |
| HXC-1239 | Feature  | P1  | Implement 10:1 DST virtual-time compression by fast-forwarding idle p                                      |
| HXC-1240 | Feature  | P1  | Achieve 1,000+ simulated nodes in a single DST process                                                     |
| HXC-1241 | Feature  | P1  | Author DST consensus+gossip workloads using SETUP→EXECUTION→CHECK→METRICS pattern                          |
| HXC-1243 | Feature  | P0  | Implement 8 network fault injectors (latency, loss, corruption, reorder, drop, partition, DNS, TCP reset)  |
| HXC-1244 | Feature  | P0  | Implement 8 node fault injectors (VM crash/restart/pause, CPU/mem/disk kill, graceful shutdown)            |
| HXC-1245 | Feature  | P1  | Implement 3 time fault injectors (clock skew, clock freeze, monotonic drift)                               |
| HXC-1247 | Feature  | P1  | Implement pure-Go in-sim chaos faults toward 25+ (ClockSkew, DiskFailure, MessageReorder, Byzantine, etc.) |
| HXC-1248 | Feature  | P0  | Build YAML chaos Scenario Engine with phases, blast radius and abort-conditions                            |

| HXC      | Type     | Pri | Title                                                                                             |
|----------|----------|-----|---------------------------------------------------------------------------------------------------|
| HXC-1249 | Feature  | P0  | Implement emergency-stop and auto-recovery with <=2s halt latency                                 |
| HXC-1254 | Feature  | P1  | Implement TestRunner with parallel suite execution and result collection                          |
| HXC-1255 | Feature  | P1  | Implement session test state machine (IDLE→SETUP→RUNNING→CHAOS_INJECT→VERIFY→RECO                 |
| HXC-1258 | Feature  | P1  | Implement MetricsCollector exporting 15+ chaos Prometheus metric ser<br>OpenTelemetry tracing     |
| HXC-1259 | Feature  | P1  | Implement HelixQA automatic challenge generation from test outcomes                               |
| HXC-1260 | Feature  | P1  | Implement metrics validation against baseline KPI table with severity ga                          |
| HXC-1261 | Feature  | P1  | Implement Welch's t-test statistical regression detector for HelixQA                              |
| HXC-1264 | Task     | P1  | Define WIT interfaces for device-simulator, workload-generator, fault-in<br>exporter              |
| HXC-1265 | Feature  | P1  | Implement capability-based plugin sandbox with resource limits and auc                            |
| HXC-1267 | Feature  | P0  | Implement cmd/helix-test CLI with dst/chaos/device/snapshot subcomm                               |
| HXC-1268 | Feature  | P1  | Build cmd/helix-snapshot standalone CLI reusing pkg/testing/snapshot                              |
| HXC-1278 | Research | P3  | Add property-based + coverage-guided fuzzing for cluster operations                               |
| HXC-1290 | Feature  | P0  | Implement pkg/device universal capability Descriptor with 15-tier + trus                          |
| HXC-1291 | Feature  | P0  | Implement device.Probe(ctx) arch/CPU/mem auto-detection with overri                               |
| HXC-1292 | Feature  | P0  | Implement device.AssignTier decision tree per §50 taxonomy (riscv64-><br>fw->T1, GPU+battery->T9) |
| HXC-1293 | Feature  | P0  | Extend api/v1/node.proto with tier, trust_level, compute_class, arch field                        |
| HXC-1294 | Feature  | P0  | Create api/v1/resources.proto with npu_tops, fpga_logic_elements, tflop                           |
| HXC-1295 | Feature  | P0  | Extend api/v1/scheduler.proto with tier constraints and power_budget                              |
| HXC-1296 | Feature  | P0  | Implement device.Descriptor <-> pb.Node round-trip mapping helpers                                |
| HXC-1297 | Feature  | P0  | Implement pkg/scheduler tier-aware + power-aware Plugin (tier_power,                              |
| HXC-1298 | Feature  | P0  | Implement power-aware T9 HANDHELD scheduling policy (battery/th<br>active gating)                 |
| HXC-1299 | Feature  | P0  | Implement pkg/discovery 15-tier registry + SelectByTier(tier, minTrust)                           |
| HXC-1300 | Feature  | P0  | Extend pkg/resources to probe NPU TOPS, FPGA logic elements, GPU                                  |
| HXC-1301 | Feature  | P1  | Implement pkg/cloudspot PreemptionWatcher with injectable SignalSou<br>hooks                      |
| HXC-1302 | Feature  | P1  | Extend pkg/wireguard with preemption-safe teardown / spot-drain hook                              |
| HXC-1303 | Feature  | P1  | Implement pkg/inference provider-agnostic Backend interface + capabil                             |
| HXC-1304 | Feature  | P1  | Implement pkg/inference LocalBackend wrapping internal/llm manager                                |
| HXC-1305 | Feature  | P1  | Implement device discovery engine: GPU detection (Vulkan/CUDA/RO<br>strategies)                   |
| HXC-1306 | Feature  | P1  | Implement device discovery engine: NPU detection (Rockchip/NVIDIA<br>Apple)                       |
| HXC-1307 | Feature  | P2  | Implement device discovery engine: FPGA detection (Xilinx/Intel/Lattic<br>core)                   |
| HXC-1308 | Feature  | P2  | Implement micro-benchmark producing normalized per-compute-class                                  |
| HXC-1309 | Feature  | P1  | Implement capability manifest generation + control-plane negotiation                              |
| HXC-1310 | Feature  | P0  |                                                                                                   |

| HXC      | Type    | Pri | Title                                                                        |
|----------|---------|-----|------------------------------------------------------------------------------|
|          |         |     | Implement per-tier security model enforcement (sandbox + data/network level) |
| HXC-1311 | Docs    | P1  | Author complete YAML tier definitions (T1-T15) with min_requirements         |
| HXC-1328 | Feature | P1  | Implement cloud-spot AWS/Azure/GCP IMDS interruption pollers (adapted)       |
| HXC-1331 | Docs    | P2  | Author 64-device master taxonomy table as machine-readable device catalog    |
| HXC-1334 | Feature | P0  | Implement pkg/hlc Hybrid Logical Clock with drift clamp and total order      |
| HXC-1335 | Feature | P0  | Implement pkg/crdt G-Counter with per-replica max-merge convergence          |
| HXC-1336 | Feature | P0  | Implement pkg/crdt LWW-Register with HLC last-writer-wins tiebreak           |
| HXC-1337 | Feature | P0  | Implement pkg/crdt OR-Set with unique-tag add/remove semantics               |
| HXC-1338 | Feature | P1  | Implement pkg/crdt/merkle anti-entropy tree with divergent-key Diff          |
| HXC-1339 | Feature | P1  | Implement Vector Clock causality type for Tier-2 operational state           |
| HXC-1340 | Feature | P0  | Implement pkg/federation Cell model and lifecycle state machine              |
| HXC-1341 | Feature | P0  | Implement thread-safe Cell Registry with add/remove/list/snapshot            |
| HXC-1342 | Feature | P1  | Implement pkg/federation federated API aggregation with partial failure      |
| HXC-1343 | Feature | P1  | Implement Phi-accrual failure detector in pkg/swim                           |
| HXC-1344 | Feature | P0  | Implement pkg/swim/hierarchical two-tier LAN+WAN delegate gossip             |
| HXC-1345 | Feature | P1  | Implement gateway-relayed cross-cell suspicion propagation                   |
| HXC-1346 | Feature | P1  | Implement bootstrap/rendezvous strategies (static, DNS-SRV, DHT, cloud)      |
| HXC-1347 | Feature | P1  | Implement mDNS/DNS-SD local cell discovery advertiser and browser            |
| HXC-1348 | Feature | P0  | Implement pkg/nat traversal STUN client and NAT-type classification          |
| HXC-1349 | Feature | P0  | Implement ICE candidate gathering, prioritization and connectivity check     |
| HXC-1350 | Feature | P1  | Implement TURN relay allocation and fallback chain                           |
| HXC-1351 | Feature | P1  | Implement QUIC fallback transport with 0-RTT and connection migration        |
| HXC-1352 | Feature | P0  | Implement internal/cell WireGuard inter-cell mesh manager                    |
| HXC-1353 | Feature | P1  | Implement WireGuard zero-downtime key rotation with overlap window           |
| HXC-1356 | Feature | P2  | Integrate libp2p DHT discovery, GossipSub and circuit relay                  |
| HXC-1357 | Feature | P1  | Implement pkg/spiffe/federation trust-bundle exchange across cells           |
| HXC-1358 | Feature | P2  | Implement OPA/Gatekeeper cross-cluster federated-trust admission             |
| HXC-1359 | Feature | P2  | Implement OPA data-residency / data-sovereignty admission policy             |
| HXC-1361 | Feature | P1  | Implement double-encryption (WireGuard L3 + mTLS L7) data path               |
| HXC-1363 | Feature | P1  | Implement federated service discovery (cell-local + global registry)         |
| HXC-1364 | Feature | P1  | Implement internal/federation Karmada PropagationPolicy engine               |
| HXC-1365 | Feature | P1  | Extend pkg/scheduler with latency-aware spot/preemptible scoring             |
| HXC-1367 | Feature | P1  | Implement pkg/gitops ArgoCD ApplicationSet federation client                 |
| HXC-1368 | Feature | P2  | Implement CRDT-based config sync for cell-local overrides                    |
| HXC-1371 | Task    | P1  | Enforce etcd-stays-within-cell static-analysis lint gate                     |
| HXC-1372 | Feature | P2  | Implement per-cell etcd Raft WAN-safe tuning profiles                        |
| HXC-1373 | Feature | P1  | Implement split-brain prevention and partition-classification logic          |
| HXC-1374 | Feature | P2  | Implement internal/chaos CE-01..CE-12 chaos experiment suite                 |
| HXC-1375 | Feature | P2  | Implement Turmoil-style deterministic multi-cluster simulation harness       |



| HXC      | Type    | Pri | Title                                                                                    |
|----------|---------|-----|------------------------------------------------------------------------------------------|
| HXC-1377 | Docs    | P2  | Author Phase 6 FMEA catalog with detection and recovery runbooks                         |
| HXC-1380 | Feature | P1  | Implement split-brain detection PromQL alerts with runbook links                         |
| HXC-1381 | Feature | P3  | Build Grafana global-health cell-status-grid dashboard                                   |
| HXC-1383 | Feature | P2  | Implement five federation topology patterns and block-binding modes                      |
| HXC-1386 | Task    | P1  | Verify sub-phase dependency gates 6a->6b->6c->6d soak conditions                         |
| HXC-1387 | Feature | P0  | Implement Multi-Raft per-shard consensus manager (G-01, P0)                              |
| HXC-1388 | Feature | P0  | Implement heartbeat coalescer for Multi-Raft network overhead control                    |
| HXC-1389 | Feature | P1  | Implement leaseholder fast-path local reads in Multi-Raft                                |
| HXC-1390 | Feature | P0  | Implement MVCC revision store with B-tree index and time-travel (G-02)                   |
| HXC-1391 | Feature | P1  | Implement persistent Watch Manager with synced/unsynced groups (G-03)                    |
| HXC-1392 | Feature | P0  | Implement SLURM-style backfill scheduler over availability timeline (G-04)               |
| HXC-1393 | Feature | P1  | Implement multifactor priority queue for scheduler (age+fairshare+size+...)              |
| HXC-1394 | Feature | P0  | Implement CRC16 16,384 hash-slot session router (G-03, P0)                               |
| HXC-1395 | Feature | P0  | Implement MOVED/ASK redirection and ASKING for in-flight slot migration                  |
| HXC-1396 | Feature | P2  | Implement Atomic Slot Migration controller for live session migration (G-05)             |
| HXC-1397 | Feature | P2  | Add config-epoch conflict resolution for simultaneous failovers                          |
| HXC-1398 | Feature | P1  | Add Startup health-probe tier with GPU grace period (G-04, P1)                           |
| HXC-1399 | Feature | P1  | Implement SWIM majority PFAIL->FAIL two-phase failure confirmation                       |
| HXC-1400 | Feature | P0  | Implement largest-subcluster-wins voting quorum (G-19, P0)                               |
| HXC-1401 | Feature | P1  | Implement STONITH fencing agents (IPMI/cloud/shared-disk) (G-19, P1)                     |
| HXC-1402 | Feature | P2  | Implement Pacemaker-style four-type constraint engine (G-20, P2)                         |
| HXC-1403 | Feature | P2  | Implement SCAN stable virtual-IP/DNS client endpoint (G-21, P2)                          |
| HXC-1404 | Feature | P1  | Implement N+K failover capacity admission control (G-22, P1)                             |
| HXC-1405 | Feature | P1  | Implement gang scheduling all-or-nothing GPU reservation (G-18, P1)                      |
| HXC-1406 | Feature | P2  | Implement topology-aware NUMA/NVLink GPU placement scoring (G-19)                        |
| HXC-1407 | Feature | P1  | Implement device-plugin / GRES fingerprinting framework (G-11/G-13)                      |
| HXC-1408 | Feature | P1  | Implement BOINC-style redundant-execution trust scorer (G-09, P1)                        |
| HXC-1409 | Feature | P2  | Implement standalone delta-state CRDT package (G-counter/PN-counter map) (G-01 part, P2) |
| HXC-1410 | Feature | P1  | Implement Cassandra 3-layer repair: hinted handoff + read repair + Merge (G-01 part, P1) |
| HXC-1411 | Feature | P2  | Implement Informer cache helixcache.Watcher (list-watch local cache) (G-06)              |
| HXC-1412 | Feature | P2  | Implement rate-limited work queue with exponential backoff (G-06, P2)                    |
| HXC-1413 | Feature | P2  | Implement API Priority & Fairness FlowSchema->PriorityLevel->Queue                       |
| HXC-1415 | Feature | P2  | Implement idempotent producer (PID + sequence) for exactly-once messaging                |
| HXC-1416 | Feature | P2  | Implement cooperative incremental rebalancing for consumers/members                      |
| HXC-1417 | Feature | P3  | Implement embedded KRaft-style Raft quorum to remove external ZooKeeper                  |
| HXC-1418 | Feature | P2  | Implement tiered cache hot(memory)/warm(NVMe)/cold(SSD) data tiering                     |
| HXC-1419 | Feature | P0  | Build deterministic simulation testing (DST) framework on Turmoil (G-15)                 |
| HXC-1420 | Feature | P0  | Implement BUGGIFY chaos macros at 25% deterministic fire rate (G-15)                     |

| HXC      | Type    | Pri | Title                                                                        |
|----------|---------|-----|------------------------------------------------------------------------------|
| HXC-1421 | Feature | P1  | Integrate Porcupine linearizability checking in CI under fault injection (   |
| HXC-1422 | Feature | P2  | Build nightly chaos pipeline (pod kill, partition, disk stall, clock skew) ( |
| HXC-1424 | Docs    | P2  | Produce hardened HelixCluster architecture diagram and component ma          |
| HXC-1425 | Docs    | P1  | Produce Phase 7 phase-by-phase 23-gap matrix as tracked deliverable le       |
| HXC-1426 | Feature | P1  | Implement pkg/chutes Chutes OpenAI-compatible API Client (non-strea          |
| HXC-1427 | Feature | P1  | Implement Chutes API client fallback-model chain on retrieable failure       |
| HXC-1428 | Feature | P1  | Implement SSE streaming decoder for Chutes chat completions                  |
| HXC-1429 | Feature | P1  | Implement Chutes API model-list and user-account/balance queries             |
| HXC-1430 | Feature | P1  | Implement model-router default-model resolution by strategy (latency/th      |
| HXC-1431 | Feature | P1  | Wire E2EE inference envelope between pkg/chutes and security/pkg/e2e         |
| HXC-1435 | Feature | P0  | Implement GraValVerifier.BatchVerify concurrent attestation with pass-r      |
| HXC-1436 | Feature | P0  | Implement GraVal VRAM-threshold (95%) gate in attestation                    |
| HXC-1439 | Feature | P0  | Define ChutesMinerConfig and ValidatorConfig data structures                 |
| HXC-1446 | Feature | P0  | Implement custom HelixGepetto dual-resource arbitration strategy             |
| HXC-1447 | Feature | P0  | Implement dual-workload capacity reservation in internal/gpu                 |
| HXC-1448 | Feature | P1  | Implement GraVal attestation hook in internal/gpu Manager                    |
| HXC-1449 | Feature | P2  | Implement MIG profile management in internal/gpu                             |
| HXC-1454 | Feature | P1  | Implement MarketplaceAdapter interface and Chutes adapter                    |
| HXC-1455 | Feature | P1  | Implement UnifiedManager.RouteWorkload concurrent pricing + compo            |
| HXC-1456 | Feature | P2  | Implement RevenueOptimizer.OptimizeAllocation greedy GPU-to-mark             |
| HXC-1458 | Feature | P2  | Implement Akash marketplace adapter                                          |
| HXC-1460 | Feature | P2  | Implement pkg/economics RewardDistributor multi-token distribution           |
| HXC-1461 | Feature | P2  | Implement RewardDistributor.GetParticipantROI and break-even trackin         |
| HXC-1469 | Feature | P1  | Wire internal/gateway to route internal AI requests through Chutes API c     |
| HXC-1470 | Feature | P1  | Replace internal/llm stub Inference with real model router (latency/throu    |
| HXC-1473 | Feature | P2  | Enable AWQ 4-bit quantization as default model format                        |
| HXC-1476 | Feature | P2  | Implement hybrid PQC TLS (X25519 + ML-KEM-768) node-to-node tr               |
| HXC-1480 | Feature | P2  | Implement carbon-aware scheduler with per-job energy metering                |
| HXC-1481 | Feature | P2  | Implement EU AI Act compliance documentation pipeline                        |
| HXC-1482 | Feature | P2  | Implement export-control tier verification (country-code KYC) at node c      |
| HXC-1483 | Feature | P1  | Extend pkg/metrics + Prometheus/Grafana with TAO/GraVal/throughpu            |
| HXC-1484 | Feature | P1  | Extend pkg/health with miner-api and GraVal DaemonSet health checks          |
| HXC-1493 | Feature | P0  | Implement GPUTier type with priority ordering (Local..Decentralized)         |
| HXC-1494 | Feature | P0  | Implement PoolManager registry with tier-ordered Allocate walk               |
| HXC-1495 | Feature | P0  | Implement WorkloadSpec/GPUDevice/GPUAllocation/PoolStatus data               |
| HXC-1496 | Feature | P0  | Implement candidate filtering (model/VRAM/cost/label selector)               |
| HXC-1497 | Feature | P1  | Implement global MaxCostPerHour budget cap on allocations                    |

| HXC      | Type     | Pri | Title                                                                     |
|----------|----------|-----|---------------------------------------------------------------------------|
| HXC-1498 | Feature  | P0  | Implement PriorityScheduler (tier then least-load then cost)              |
| HXC-1499 | Feature  | P1  | Implement CostAwareScheduler selecting cheapest GPUs meeting SLA          |
| HXC-1500 | Feature  | P1  | Implement LatencyAwareScheduler for inference routing                     |
| HXC-1501 | Feature  | P1  | Implement HealthMonitor with 30s checks and auto-failover trigger         |
| HXC-1502 | Feature  | P0  | Implement internal/costbroker ComputeBroker weighted scorer               |
| HXC-1503 | Feature  | P1  | Implement ComputeBroker 60s re-scoring loop with price-spike guard        |
| HXC-1504 | Feature  | P0  | Implement pkg/burst BurstController MONITOR->SPILL->RECOVER               |
| HXC-1505 | Feature  | P1  | Implement BurstController utilization RingBuffer moving average           |
| HXC-1506 | Feature  | P1  | Implement BurstController activateBurst capacity estimation + allocation  |
| HXC-1507 | Feature  | P1  | Implement pkg/burst CostRouter per-workload-type provider scoring         |
| HXC-1508 | Feature  | P1  | Implement 5-tier fallback chain Chutes->io.net->RunPod->AWS               |
| HXC-1509 | Feature  | P2  | Implement QoS tiers (real-time, interactive, batch, best-effort)          |
| HXC-1510 | Feature  | P0  | Implement pkg/provider/chutes ChutesProvider OpenAI-compatible ada        |
| HXC-1511 | Feature  | P1  | Implement Chutes retryWithFallback across fallback models on 429          |
| HXC-1512 | Feature  | P2  | Implement Chutes GetBalance USD balance monitor                           |
| HXC-1513 | Feature  | P2  | Implement Chutes streaming (SSE) chat-completion variant                  |
| HXC-1514 | Feature  | P1  | Implement pkg/provider/ionet IONetProvider Ray cluster adapter            |
| HXC-1515 | Feature  | P1  | Implement pkg/provider/runpod RunPodProvider serverless + warm pool       |
| HXC-1516 | Feature  | P2  | Implement pkg/provider/aws AWSProvider EC2 Spot adapter                   |
| HXC-1519 | Feature  | P0  | Implement pkg/local LocalGPURegistrar with TCO effective-cost             |
| HXC-1523 | Feature  | P1  | Implement pkg/e2ee GraValVerifier provider admission gate                 |
| HXC-1527 | Feature  | P1  | Define gpu_proxy.proto gRPC GPUProvider kernel-dispatch service           |
| HXC-1529 | Feature  | P1  | Implement workload-suitability classifier (HPC local vs inference remote) |
| HXC-1530 | Feature  | P0  | Implement cmd/gpu-pool-manager HTTP+gRPC front-end binary                 |
| HXC-1531 | Feature  | P0  | Implement cmd/burst-controller utilization-driven binary                  |
| HXC-1532 | Feature  | P0  | Implement cmd/e2ee-proxy transparent ML-KEM-768 proxy binary              |
| HXC-1533 | Task     | P1  | Add ProviderAdapter registration hooks to pkg/gpu                         |
| HXC-1534 | Feature  | P1  | Add GPURTier-aware filter predicate to pkg/scheduler                      |
| HXC-1535 | Feature  | P1  | Add GPU-tier utilization/cost/provider-health metrics to pkg/metrics      |
| HXC-1538 | Feature  | P2  | Implement @helix.task Go decorator/SDK with lifecycle hooks               |
| HXC-1539 | Feature  | P2  | Implement intelligent ModelRouter (latency/throughput/cost/tee/balance)   |
| HXC-1544 | Research | P3  | Implement predictive scaling forecast to pre-warm before peaks            |
| HXC-1547 | Feature  | P2  | Implement TCO calculator for hybrid own+burst economic model              |
| HXC-1548 | Feature  | P1  | Implement CostTracker monthly cost report vs AWS on-demand                |
| HXC-1551 | Task     | P1  | Build Prometheus/Grafana Phase 8B tier-utilization + cost dashboards      |
| HXC-1556 | Task     | P1  | Benchmark ML-KEM-768 E2EE handshake latency (<1ms target)                 |
| HXC-1561 | Feature  | P0  | Add ChaCha20-Poly1305 AEAD cipher variant to pkg/security/e2ee            |
| HXC-1563 | Feature  | P1  | Implement bidirectional per-request response keypair in pkg/security/e2   |
| HXC-1564 | Feature  | P2  | Implement gzip-before-encrypt payload compression in pkg/security/e2      |

| HXC      | Type    | Pri | Title                                                                                   |
|----------|---------|-----|-----------------------------------------------------------------------------------------|
| HXC-1565 | Feature | P1  | Implement streaming SSE E2EE decryption (e2e_init + per-chunk) in pkg/gpuattest         |
| HXC-1567 | Feature | P1  | Maintain E2EE length-prefixed framed transport over io.ReadWriter                       |
| HXC-1568 | Feature | P0  | Implement pkg/gpuattest device-info challenge/response + fingerprint                    |
| HXC-1569 | Feature | P0  | Implement seeded matmul proof-of-GPU-work (PoVW) in pkg/gpuattest                       |
| HXC-1570 | Feature | P0  | Implement O(1) spot-check verification in pkg/gpuattest                                 |
| HXC-1571 | Feature | P0  | Implement device-sealed encrypt/decrypt (SealForDevice/OpenFromDevice) in pkg/gpuattest |
| HXC-1572 | Feature | P2  | Implement filesystem residency challenge in pkg/gpuattest                               |
| HXC-1573 | Feature | P2  | Implement multi-GPU node enumeration in pkg/gpuattest                                   |
| HXC-1579 | Feature | P2  | Implement pkg/security/admission OPA-style policy + signed-image (cosign) support       |
| HXC-1580 | Feature | P1  | Implement pkg/modelintegrity hf_cache_verify gate (SHA-256 + size)                      |
| HXC-1581 | Feature | P2  | Implement model/revision anti-cheat verification token (V2 HMAC-SHA256)                 |
| HXC-1582 | Feature | P3  | Implement X25519 ephemeral session-key handshake for verification to model              |
| HXC-1583 | Feature | P1  | Maintain pkg/scheduler cost-aware GPU placement plugin                                  |
| HXC-1584 | Feature | P1  | Implement value-multiplier preemption in pkg/scheduler (Gepetto mode)                   |
| HXC-1585 | Feature | P1  | Implement SKIP LOCKED-style optimistic work-claiming in pkg/scheduler                   |
| HXC-1586 | Feature | P1  | Implement attestation-gated scheduler admission predicate                               |
| HXC-1587 | Feature | P2  | Implement bounty/auction economic-placement plugin for scheduler                        |
| HXC-1588 | Feature | P2  | Implement utilization-aware EWMA candidate ranking for scheduler/orchestrator           |
| HXC-1589 | Feature | P2  | Implement Kueue-style suspend-then-admit tiered job admission                           |
| HXC-1590 | Feature | P1  | Implement NodeSelector placement-constraint schema in pkg/resources                     |
| HXC-1592 | Feature | P1  | Implement SUPPORTED_GPUS compute-multiplier catalog + LookupMux                         |
| HXC-1595 | Feature | P1  | Implement streaming-safe failover (retry only before first byte) in LLMOrchestrator     |
| HXC-1596 | Feature | P2  | Implement per-task ordered fallback chains with dedup/cap + empty-responses             |
| HXC-1597 | Feature | P2  | Implement thermal pre-warm / scale-from-zero state machine (Therm)                      |
| HXC-1598 | Feature | P2  | Implement passthrough cords + disconnect-aware upstream abort in LLMOrchestrator        |
| HXC-1599 | Feature | P3  | Implement Claude/Responses API-shape adapters for LLMOrchestrator                       |
| HXC-1600 | Feature | P3  | Implement error-classified LLM failover taxonomy + sandbox scheduling                   |
| HXC-1601 | Feature | P2  | Implement pkg/inferenceproxy correlation-ID + backend audit trail                       |
| HXC-1602 | Feature | P2  | Implement deterministic keyed-hash anonymization in pkg/inferenceproxy                  |
| HXC-1603 | Feature | P2  | Implement spoof-proof managed-header sanitization at the recording edge                 |
| HXC-1604 | Feature | P1  | Maintain pkg/fiber length-prefixed framed miner<->validator transport                   |
| HXC-1605 | Feature | P1  | Implement fiber ed25519 signed-identity handshake + stake-gated admission               |
| HXC-1606 | Feature | P2  | Implement node identity verify-then-pin (TOFU) keypair-rooted TLS                       |
| HXC-1607 | Feature | P2  | Implement pkg/marketplace registration -> bounty -> metering -> payout                  |
| HXC-1608 | Feature | P2  | Implement pkg/marketplace/audit commit-then-prove reproducible recording                |
| HXC-1609 | Feature | P3  | Implement scale-to-zero metered hot/cold billing state machine                          |
| HXC-1610 | Feature | P3  | Implement watchtower-style liveness/integrity prober for served instances               |
| HXC-1612 | Feature | P3  | Implement OpenAPI request-validation admission middleware for contracts                 |
| HXC-1613 | Docs    | P1  | Document Phase 8C exit-gate evidence matrix (CLAUDE-1 usability pr                      |

| HXC      | Type    | Pri | Title                                                                                                                   |
|----------|---------|-----|-------------------------------------------------------------------------------------------------------------------------|
| HXC-1614 | Docs    | P2  | Document Phase 8C architecture integration diagrams (scheduler<-attese2ee<-TEE, marketplace)                            |
| HXC-1616 | Bug     | P2  | Honor WithClock in cloudspot Azure/GCP pollers (currently silently ignore)                                              |
| HXC-1621 | Bug     | P1  | SELinux bind-mount relabel hardcoded :Z — make conditional + cross-arch (containers/pkg/crossbuild)                     |
| HXC-1626 | Feature | P2  | HXC-1119 follow-up: wire pkg/events Avro codec into the NATS Helix + schemas for SessionEvent/SchedulerEvent/AuditEvent |
| HXC-1627 | Feature | P2  | HXC-1626 follow-up: NATSBackend typed Avro publish/subscribe for SessionEvent/SchedulerEvent/AuditEvent                 |
| HXC-1628 | Task    | P2  | buf lint STYLE conformance for api/v1 protos (PACKAGE_DIRECTORIES RPC-response naming)                                  |
| HXC-1629 | Task    | P2  | Wire Makefile migrate-up/migrate-down to scripts/run-migrations.sh (P1)                                                 |
| HXC-1630 | Task    | P1  | Run govulncheck vulnerability scan across main/api/v1/security module (gap)                                             |
| HXC-1631 | Bug     | P1  | govulncheck: bump Go toolchain (go.mod go-directive) to clear 9 reachable advisories                                    |
| HXC-1632 | Bug     | P2  | govulncheck: bump golang.org/x/net v0.54.0 -> v0.55.0 (GO-2026-5026)                                                    |
| HXC-1633 | Feature | P1  | Implement real wireguard-go userspace path on macOS (CLAUDE-2 patch wireguard)                                          |
| HXC-1634 | Feature | P2  | Implement real darwin GPU/device probes (Metal/IOKit) for internal/gpu (CLAUDE-2)                                       |
| HXC-1635 | Task    | P2  | Add SBOM generation + dependabot/renovate to complete supply-chain items 34/80)                                         |
| HXC-1636 | Task    | P2  | Local make deps-update target (no-CI dependency-update maintenance, P1)                                                 |
| HXC-1637 | Bug     | P1  | make build is broken: builds non-existent ./cmd/helix-cluster                                                           |
| HXC-1638 | Bug     | P2  | helixctl binary referenced in docs but absent from cmd/                                                                 |
| HXC-1639 | Bug     | P2  | SQL schema drift: 0001_primary_schema.sql diverges from the golang- (001-015)                                           |
| HXC-1640 | Feature | P1  | pkg/stonith fencing-safety test (positive confirmation + fenced node cannot)                                            |
| HXC-1641 | Bug     | P1  | FIX registry NextHXCID lexical-vs-numeric id-allocation bug                                                             |
| HXC-1642 | Feature | P1  | pkg/epochresolve adversarial fencing test (stale epoch rejected, at-most-one)                                           |
| HXC-1643 | Bug     | P0  | FIX pkg/idempotent concurrency bug (data race + double-commit broke)                                                    |
| HXC-1644 | Feature | P1  | pkg/workclaim concurrent mutual-exclusion test (exactly-one claimant)                                                   |
| HXC-1645 | Feature | P1  | pkg/voting exhaustive at-most-one-authoritative test (split-brain invariant)                                            |
| HXC-1646 | Feature | P1  | pkg/semaphore adversarial bounded-concurrency test (never exceed N, no)                                                 |
| HXC-1647 | Feature | P1  | pkg/lru adversarial eviction + capacity test (true LRU not FIFO, exact bound)                                           |
| HXC-1648 | Bug     | P0  | FIX pkg/lock double-release crash + cross-owner free (unguarded Unlock)                                                 |
| HXC-1649 | Bug     | P0  | FIX pkg/heartbeatcoalescer Receiver data race (unsynchronized lastSeen)                                                 |
| HXC-1650 | Bug     | P1  | harden EtdLocker fire-once release guard (follow-up to HXC-1648)                                                        |
| HXC-1651 | Feature | P1  | pkg/flowcontrol concurrent credit-conservation test (never over-admit)                                                  |
| HXC-1652 | Feature | P1  | pkg/ratelimit UserLimiter adversarial over-admission test (dual-bucket, P1)                                             |
| HXC-1653 | Feature | P1  |                                                                                                                         |

| HXC      | Type    | Pri | Title                                                                                     |
|----------|---------|-----|-------------------------------------------------------------------------------------------|
|          |         |     | pkg/pool GPU-instance-pool adversarial concurrency test (no double-iss bound)             |
| HXC-1654 | Feature | P1  | pkg/watchmanager adversarial concurrency test (register/notify/sync chu                   |
| HXC-1655 | Feature | P1  | pkg/multiraft adversarial registry-concurrency test (concurrent create/ro                 |
| HXC-1656 | Feature | P1  | pkg/pubsub adversarial concurrent sub/unsub/publish test (no send-on-c                    |
| HXC-1657 | Feature | P1  | pkg/tieredcache coherence test (re-Put shadows stale lower tier, no conta                 |
| HXC-1658 | Feature | P1  | pkg/session adversarial concurrent FSM test (at-most-once terminal tran                   |
| HXC-1659 | Bug     | P1  | FIX pkg/offlinesync Reconcile conflict winner non-deterministic (violat lower-Seq policy) |
| HXC-1660 | Feature | P1  | pkg/workqueue adversarial concurrency test (no double-processing, ded                     |
| HXC-1661 | Feature | P1  | pkg/metering conservation test (exact revenue accounting, per-key isolat                  |
| HXC-1662 | Feature | P1  | pkg/budgetcap concurrent over-spend test (cap never exceeded)                             |
| HXC-1663 | Feature | P1  | pkg/billingfsm adversarial money-safety test (transition matrix, no doub conservation)    |
| HXC-1664 | Feature | P1  | pkg/informer cache-coherence test (latest-delta-wins, delete coherence, r                 |
| HXC-1665 | Feature | P1  | pkg/rebalance adversarial minimal-disruption test (sticky, balance, idem                  |
| HXC-1666 | Bug     | P0  | FIX pkg/slotmigration concurrent-map crash (routing lookups raced the                     |
| HXC-1667 | Bug     | P0  | FIX pkg/edgeregistry concurrent-map race (Register raced Get/Len/List                     |
| HXC-1668 | Feature | P1  | pkg/modelrouter concurrent read-safety test (immutable routing table)                     |
| HXC-1669 | Feature | P1  | pkg/fedtopology concurrent read-safety + path-correctness test (immutab                   |
| HXC-1670 | Bug     | P0  | FIX pkg/jwt keystore concurrent-map crash on auth path (verify raced ro                   |
| HXC-1671 | Feature | P1  | pkg/ewmarank concurrency-contract + ranking-correctness test (docume threaded)            |
| HXC-1672 | Bug     | P0  | FIX pkg/jobadmit concurrent-map crash (admit/complete raced state loo                     |
| HXC-1673 | Feature | P1  | pkg/failconfirm concurrency-contract + quorum-confirmation test (docu threaded)           |
| HXC-1674 | Bug     | P1  | FIX pkg/redundantexec concurrent-map + trust-field race (validate race                    |
| HXC-1675 | Bug     | P1  | FIX pkg/rescorer concurrent ranking/map race (Tick rerank raced Top/R                     |
| HXC-1676 | Feature | P1  | pkg/fedtrust concurrent trust-policy test (immutable-after-build, deny-ne                 |
| HXC-1677 | Bug     | P1  | FIX pkg/thermalwarm concurrent-map + state-field race (dispatch/state                     |
| HXC-1678 | Feature | P1  | pkg/phasegate concurrent read-safety + gating-correctness test (immutab                   |
| HXC-1679 | Feature | P1  | pkg/dataplane partial-lock audit + concurrent race test (Distributor.pendi                |
| HXC-1680 | Feature | P1  | pkg/inferenceproxy partial-lock audit + concurrent race test (MemAudit                    |
| HXC-1681 | Feature | P1  | pkg/timefault concurrency-contract + fault-model test (deterministic sing harness)        |
| HXC-1682 | Feature | P1  | pkg/auctionplace adversarial clearing test (first-caller-wins, order-indepe winner)       |
| HXC-1683 | Feature | P1  | pkg/constraints Forbidden hard-ban precedence test (closes latent Forbid gap)             |
| HXC-1684 | Feature | P1  | pkg/preempt 20k-case oracle preemption test (priority-safety, minimality                  |
| HXC-1685 | Feature | P1  | pkg/carbonsched oracle selection test (lowest-carbon argmin, order-inde                   |

| HXC      | Type    | Pri | Title                                                                                 |
|----------|---------|-----|---------------------------------------------------------------------------------------|
| HXC-1686 | Feature | P1  | pkg/costsched oracle selection test (lowest-cost, lex tie-break; closes uns           |
| HXC-1687 | Feature | P1  | pkg/suitability exhaustive placement test (HPC never remote/non-low-la                |
| HXC-1688 | Feature | P1  | pkg/revenueopt adversarial pricing test (TEE->Chutes bonus, oracle win                |
| HXC-1689 | Feature | P1  | pkg/forecast numeric OLS-correctness test (formula vs oracle, exact win               |
| HXC-1690 | Bug     | P1  | FIX pkg/qos intransitive near-tie comparator (order-dependent QoS sele                |
| HXC-1691 | Feature | P1  | pkg/latencysched comparator transitivity audit + order-independence tes<br>band)      |
| HXC-1692 | Feature | P1  | pkg/gpupool comparator transitivity audit + order-independence test (tra              |
| HXC-1693 | Feature | P1  | pkg/smartrouter dual audit (comparator transitivity + partial-lock) clean             |
| HXC-1694 | Feature | P1  | pkg/bursthysteresis exhaustive hysteresis-boundary test (no flap, inclusi             |
| HXC-1695 | Feature | P1  | pkg/healthmonitor FSM transition-matrix + concurrency test (edge-trigg<br>guarded)    |
| HXC-1696 | Feature | P1  | pkg/balancemonitor event-log conservation + concurrency test (no lost a<br>threshold) |
| HXC-1697 | Feature | P1  | pkg/hybridkex adversarial hybrid-binding test (X25519+ML-KEM-768,<br>bind)            |
| HXC-1698 | Feature | P1  | pkg/x25519session adversarial key-agreement test (auth, wrong-peer, lo                |
| HXC-1699 | Feature | P1  | pkg/modelintegrity adversarial tamper/forgery test (SHA-256 content-pi                |
| HXC-1700 | Feature | P1  | pkg/gpuattest adversarial attestation-forgery test (ed25519, pinned-key t             |
| HXC-1701 | Feature | P1  | pkg/spiffefed adversarial trust-domain-confusion test (exact host match,<br>bypass)   |
| HXC-1702 | Task    | P1  | attestadmit fail-closed adversarial tests                                             |
| HXC-1703 | Task    | P1  | gravalverify anti-cheat adversarial tests                                             |
| HXC-1704 | Task    | P1  | exportcontrol embargo/lookalike adversarial tests                                     |
| HXC-1705 | Task    | P1  | imagepolicy admission adversarial tests                                               |
| HXC-1706 | Task    | P1  | headersanitize spoof-proofing adversarial tests                                       |
| HXC-1707 | Task    | P1  | edgeverify content-hash adversarial tests                                             |
| HXC-1708 | Task    | P1  | anonymize keyed-hash tokenizer adversarial tests                                      |
| HXC-1709 | Task    | P1  | priorityqueue total-order adversarial tests                                           |
| HXC-1710 | Task    | P1  | auditproof canonical-serialization adversarial tests                                  |
| HXC-1711 | Task    | P1  | backoff cap/overflow adversarial tests                                                |
| HXC-1712 | Task    | P1  | nodeselector constraint-satisfaction adversarial tests                                |
| HXC-1713 | Task    | P1  | economics conservation/ROI adversarial tests                                          |
| HXC-1714 | Task    | P1  | hybridtco monotonicity/accounting adversarial tests                                   |
| HXC-1715 | Task    | P1  | costrouter cost-optimal selection adversarial tests                                   |
| HXC-1716 | Task    | P1  | fallbackchain orchestration adversarial tests                                         |
| HXC-1717 | Task    | P1  | openapivalidate fail-closed validation adversarial tests                              |
| HXC-1718 | Task    | P1  | correlation audit-trail adversarial tests                                             |
| HXC-1719 | Task    | P1  | modelretry classification adversarial tests                                           |
| HXC-1720 | Task    | P1  | workloadrouter routing+concurrency adversarial tests                                  |
| HXC-1721 | Task    | P1  | streamfailover orchestration adversarial tests                                        |

| HXC      | Type | Pri | Title                                                                     |
|----------|------|-----|---------------------------------------------------------------------------|
| HXC-1722 | Task | P1  | burstcapacity spillover-estimator adversarial tests                       |
| HXC-1723 | Task | P1  | powergater night-window/state adversarial tests                           |
| HXC-1724 | Task | P1  | residency data-residency compliance adversarial tests                     |
| HXC-1725 | Task | P1  | watchtower tamper-integrity adversarial tests                             |
| HXC-1726 | Task | P1  | fsresidency file content-integrity adversarial tests                      |
| HXC-1727 | Task | P1  | devicecatalog lookup adversarial tests                                    |
| HXC-1728 | Task | P1  | gputopo topology-selection adversarial tests                              |
| HXC-1729 | Task | P1  | configsnc CRDT store concurrency adversarial tests                        |
| HXC-1730 | Task | P1  | config defaults/validation adversarial tests                              |
| HXC-1731 | Task | P1  | deviceprofile YAML-load adversarial tests                                 |
| HXC-1732 | Task | P1  | gpucatalog scoring/multiplier adversarial tests                           |
| HXC-1733 | Task | P1  | healthprobe grace-window adversarial tests                                |
| HXC-1734 | Task | P1  | devicemap round-trip mapping adversarial tests                            |
| HXC-1735 | Task | P1  | burst autoscaling-hysteresis adversarial tests                            |
| HXC-1736 | Task | P1  | archlint doc-vs-disk linter adversarial tests                             |
| HXC-1737 | Task | P1  | etcdlint import-ban linter adversarial tests                              |
| HXC-1738 | Task | P1  | tierdetect per-OS detection adversarial tests                             |
| HXC-1739 | Task | P1  | llmadapter translation-fidelity adversarial tests                         |
| HXC-1740 | Task | P1  | passthrough streaming-fidelity adversarial tests                          |
| HXC-1741 | Task | P1  | grpcutil auth-interceptor adversarial tests                               |
| HXC-1742 | Task | P1  | middleware panic-recovery/chain adversarial tests                         |
| HXC-1743 | Task | P1  | llmfailover failover-path/hint adversarial tests                          |
| HXC-1744 | Task | P1  | capability manifest/negotiate adversarial tests                           |
| HXC-1745 | Task | P1  | sandbox capability-isolation adversarial tests                            |
| HXC-1746 | Task | P1  | splitbrainalert detection adversarial tests                               |
| HXC-1747 | Task | P1  | serde capnp serialization adversarial tests                               |
| HXC-1748 | Task | P1  | netutil SSRF-defense adversarial tests                                    |
| HXC-1749 | Task | P1  | healthmonitor FSM/concurrency adversarial tests                           |
| HXC-1750 | Task | P1  | leader fencing split-brain-safety adversarial tests                       |
| HXC-1751 | Task | P1  | workclaim claim-exclusivity adversarial tests                             |
| HXC-1752 | Task | P1  | tierdef config-load adversarial tests                                     |
| HXC-1753 | Bug  | P0  | hlc Update/Now logical-counter overflow breaks HLC domination (caus       |
| HXC-1754 | Task | P1  | hashslot CRC16/hashtag adversarial tests                                  |
| HXC-1755 | Task | P1  | mvcc snapshot-isolation adversarial tests                                 |
| HXC-1756 | Task | P1  | stats Welch-t-test adversarial tests                                      |
| HXC-1757 | Task | P1  | ringavg moving-average adversarial tests                                  |
| HXC-1758 | Task | P1  | crdt semilattice-law adversarial tests                                    |
| HXC-1759 | Bug  | P1  | antientropy equal-version conflict never converges (permanent split-brain |
| HXC-1760 | Task | P1  | splitbrain quorum-classification adversarial tests                        |



| HXC      | Type | Pri | Title                                                                                           |
|----------|------|-----|-------------------------------------------------------------------------------------------------|
| HXC-1761 | Task | P1  | porcupine linearizability-checker adversarial tests                                             |
| HXC-1762 | Bug  | P1  | offlinesync: decompression-bomb DoS + equal-Seq split-brain — BOTH                              |
| HXC-1763 | Task | P1  | heartbeatcoalescer coalescing/concurrency adversarial tests                                     |
| HXC-1764 | Task | P1  | multiraft durability-safety adversarial tests                                                   |
| HXC-1765 | Bug  | P2  | computeproto: FlatBuffers decode DoS (panic on hostile bytes) — hardened<br>ReadComputeTaskSafe |
| HXC-1766 | Task | P1  | nattraversal STUN codec adversarial tests                                                       |
| HXC-1767 | Task | P1  | websocket envelope codec adversarial tests                                                      |
| HXC-1768 | Task | P1  | fiber framing/admission adversarial tests                                                       |
| HXC-1769 | Task | P1  | ice RFC8445 priority adversarial tests                                                          |
| HXC-1770 | Task | P1  | wasm sandbox-enforcement adversarial tests                                                      |
| HXC-1771 | Bug  | P1  | helixnet prod fabric teardown race: send on closed channel panic — FIXED                        |
| HXC-1772 | Task | P1  | edgefusion windowed-fusion adversarial tests                                                    |
| HXC-1773 | Task | P1  | watchmanager watch-fanout concurrency adversarial tests                                         |
| HXC-1774 | Task | P1  | informer cache-coherence adversarial tests                                                      |
| HXC-1775 | Task | P1  | cellmesh overlap-detection adversarial tests                                                    |
| HXC-1776 | Task | P1  | edgeheartbeat liveness/concurrency adversarial tests                                            |
| HXC-1777 | Task | P1  | checkpoint_merge merge/migrate adversarial tests                                                |
| HXC-1778 | Task | P1  | kraft metadata-log determinism adversarial tests                                                |
| HXC-1779 | Task | P1  | gepetto reservation/high-water adversarial tests                                                |
| HXC-1780 | Task | P1  | edgeregistry concurrency/copy-isolation adversarial tests                                       |
| HXC-1781 | Task | P1  | marketplace/scan/chutes adversarial tests (over-commit, routing, attestations)                  |
| HXC-1782 | Bug  | P1  | classads parser/evaluator stack-overflow DoS on deeply-nested expressions                       |
| HXC-1783 | Task | P1  | raftprofile parse/timing-invariant adversarial tests                                            |
| HXC-1784 | Task | P1  | phasegate gating/concurrency adversarial tests                                                  |
| HXC-1785 | Bug  | P1  | costtracker unsynchronized accumulator: data race + cost non-conservative                       |
| HXC-1786 | Task | P1  | providerchain fallback adversarial tests                                                        |
| HXC-1787 | Task | P1  | modelrouter selection adversarial tests                                                         |
| HXC-1788 | Bug  | P1  | dataplane Distributor.SendToWorker concurrent-send libzmq SIGABRT                               |
| HXC-1789 | Task | P1  | marketplaceadapter translation-fidelity adversarial tests                                       |
| HXC-1790 | Task | P1  | raftleader election-safety adversarial tests                                                    |
| HXC-1791 | Bug  | P1  | budgetcap: NaN cost bypasses MaxCostPerHour cap (unbounded over-spend)                          |
| HXC-1792 | Task | P3  | metering: adversarial test pinning documented single-goroutine contract                         |
| HXC-1793 | Task | P3  | inferenceproxy: adversarial concurrency/edge hardening test                                     |
| HXC-1794 | Bug  | P1  | ewmarank: NaN sample poisons EWMA and breaks Rank total order (memory)                          |
| HXC-1795 | Bug  | P1  | admissioncontrol: int64 reserve-sum overflow bypasses N+K capacity constraint                   |
| HXC-1796 | Task | P3  | auctionplace: adversarial test, FCFS board contract pinned (no bug)                             |
| HXC-1797 | Task | P3  | flowcontrol: adversarial test, APF round-robin cursor survives resort (no bug)                  |
| HXC-1798 | Bug  | P2  | forecast: non-finite sample poisons OLS prediction and suppresses PreWarmup<br>scaling)         |

| HXC      | Type | Pri | Title                                                                       |
|----------|------|-----|-----------------------------------------------------------------------------|
| HXC-1799 | Task | P3  | gravaladmit: adversarial test, single-use nonce gate K=1 over-commit-pr     |
| HXC-1800 | Task | P3  | balancemonitor: adversarial test, wire NaN/Inf rejected by stdlib JSON (    |
| HXC-1801 | Task | P3  | fedtrust: adversarial test, deny branches fail-CLOSED (no bug)              |
| HXC-1802 | Bug  | P2  | carbonsched: NaN carbon-intensity poisons greenest-region selection +       |
| HXC-1803 | Task | P3  | constraints: adversarial test, flat-struct placement engine (no bug)        |
| HXC-1804 | Task | P3  | federation: adversarial test, uint32+enum total order, mutex-guarded (no    |
| HXC-1805 | Task | P3  | discovery: adversarial test, all registries promise+keep concurrency safe   |
| HXC-1806 | Task | P3  | cloudspot: adversarial test, preemption-drain handler deterministic (no b   |
| HXC-1807 | Task | P3  | gpupool: adversarial test, stateless read-only query pkg (no bug)           |
| HXC-1808 | Task | P3  | backfill: adversarial test, pure deterministic scheduler (no bug)           |
| HXC-1809 | Task | P3  | health: adversarial test, fail-closed rollup, RWMutex-guarded (no bug)      |
| HXC-1810 | Bug  | P2  | rescorer: NaN price breaks rerank comparator -> NaN provider seizes to      |
| HXC-1811 | Task | P3  | idempotent: adversarial test, exactly-once under concurrent first-use (no   |
| HXC-1812 | Task | P3  | events: adversarial test, handler-func Bus race-free, no teardown-race su   |
| HXC-1813 | Task | P3  | anticheat: adversarial test, HMAC token fail-CLOSED on all malformed        |
| HXC-1814 | Bug  | P1  | edge: Allowed() admits invalid/unknown TrustLevel to all workloads (fa      |
| HXC-1815 | Task | P3  | failconfirm: adversarial test, strict-majority quorum + reporter dedup cor  |
| HXC-1816 | Task | P3  | etcd: adversarial test, thin clientv3 wrapper + key-scheme pins (no bug)    |
| HXC-1817 | Task | P3  | deviceplugin: adversarial test, Registry no-double-alloc + conservation (   |
| HXC-1818 | Bug  | P2  | gitops: PlanRollingSync silently drops apps with unknown/typo/empty t       |
| HXC-1819 | Bug  | P2  | hxcregistry: CreateItem/UpdateItem non-atomic -> orphan item rows wi        |
| HXC-1820 | Task | P3  | fmea: adversarial test, int-only RPN scoring (no bug)                       |
| HXC-1821 | Task | P3  | gpu: adversarial test, provider-adapter registry RWMutex-guarded (no b      |
| HXC-1822 | Bug  | P2  | jobadmit: used+Request int overflow bypasses quota cap (over-admit +        |
| HXC-1823 | Bug  | P2  | smartrouter: NaN/Inf score latched at index 0 mis-routes (breaks all 4 st   |
| HXC-1824 | Task | P3  | voting: adversarial test, strict-majority quorum + dedup + deterministic    |
| HXC-1825 | Bug  | P3  | jwt: verifyRSA passes crypto.Hash(0) not SHA256/384/512 -> rejects al       |
| HXC-1826 | Bug  | P1  | security: IsValidTrustDomain admits '/' in SPIFFE trust domain (fail-op     |
| HXC-1827 | Bug  | P2  | storage: RedisStore.SetSessionRouting non-atomic -> orphan routing ha       |
| HXC-1828 | Bug  | P2  | validator: checkMin/checkMax float NaN bypasses range bounds (fail-o        |
| HXC-1829 | Task | P3  | stonith: adversarial test, fencing confirms sink-side (no fail-open) (no bu |
| HXC-1830 | Task | P3  | metrics: adversarial test, atomic counters + mutex histograms conserve u    |
| HXC-1831 | Task | P3  | session: adversarial test, Manager RWMutex-guarded + deep-copy Get -        |
| HXC-1832 | Bug  | P2  | retry: exponential/linear backoff overflows time.Duration -> negative de    |

| HXC      | Type | Pri | Title                                                                                                         |
|----------|------|-----|---------------------------------------------------------------------------------------------------------------|
| HXC-1833 | Task | P3  | crypto: adversarial test, Ed25519/AES-GCM/PBKDF2 wrapper rejects t                                            |
| HXC-1834 | Task | P3  | workerpool: adversarial test, bounded pool correct (no bug)                                                   |
| HXC-1835 | Bug  | P1  | tiersec: tierRank Sscanf admits malformed tier names -> fail-open cleara                                      |
| HXC-1836 | Task | P3  | tieredcache: adversarial test, LRU bound/eviction/recency correct (no bu                                      |
| HXC-1837 | Task | P3  | doublecrypt: adversarial test, mTLS-over-WireGuard tamper/auth enforc                                         |
| HXC-1838 | Task | P3  | covgate: adversarial test, threshold >= boundary + NaN fail-closed (no b                                      |
| HXC-1839 | Bug  | P1  | qualitygate: NaN metric/threshold records no breach -> CRITICAL gate<br>poisoned metric (fail-open, CLAUDE-1) |
| HXC-1840 | Bug  | P2  | inference: 2 stale tests fail on main (encode retired internal/llm synthetic<br>RED                           |
| HXC-1841 | Task | P3  | pool: adversarial test, mutex-guarded pool no over-commit/double-hand                                         |
| HXC-1842 | Task | P3  | quantization: adversarial test, int-only VRAM estimate (no bug)                                               |
| HXC-1843 | Task | P3  | raft: adversarial test, hashicorp/raft wrapper + local FSM safety pinned (                                    |
| HXC-1844 | Task | P3  | device: adversarial test, deterministic classifier + real per-OS probe, CL<br>(no bug)                        |
| HXC-1845 | Task | P3  | dst: adversarial test, deterministic-simulation harness is genuinely deter                                    |
| HXC-1846 | Task | P3  | provider: adversarial test, pure HTTP adapter no selector/registry (no bu                                     |
| HXC-1847 | Bug  | P2  | timefault: abs64(MinInt64) overflow misses max-skew detection -> gran<br>clock (split-brain)                  |
| HXC-1848 | Task | P3  | devicemap: adversarial concurrency test, pure stateless conversion (no b                                      |
| HXC-1849 | Task | P3  | fedtopology: adversarial test, deterministic immutable topology (no bug)                                      |
| HXC-1850 | Task | P3  | thermalwarm: adversarial test, duration-based warm-up boundary correc                                         |
| HXC-1851 | Bug  | P2  | tracing: Span.Duration() ignores endTime -> finished span duration grov<br>(wrong latency)                    |
| HXC-1852 | Bug  | P2  | agentprovision: ModeManager.Submit no duplicate-ID guard -> double-<br>live-job eviction                      |
| HXC-1853 | Task | P3  | chaosexp: adversarial test, conservation + determinism (no bug)                                               |
| HXC-1854 | Task | P3  | errors: adversarial test, Is/As wrap-chain + Wrap(nil)==nil (no bug)                                          |
| HXC-1855 | Bug  | P1  | swim: handleDead lacks self-target guard -> node accepts death rumour<br>denial of liveness)                  |
| HXC-1856 | Bug  | P2  | build: Job.clone() aliases BuildArgs map despite documented deep-copy<br>mutation corrupts canonical job      |
| HXC-1857 | Task | P3  | log: adversarial test, slog shared-mutex concurrent integrity (no bug)                                        |
| HXC-1858 | Task | P3  | bootstrap: adversarial test, peer-rendezvous deterministic + no fail-open                                     |
| HXC-1859 | Task | P3  | wireguard: adversarial test, CLAUDE-2 darwin parity REAL (wireguard<br>race-clean (no bug)                    |
| HXC-1860 | Task | P3  | grafanadash: adversarial test, deterministic JSON + escaping safe (no bu                                      |
| HXC-1861 | Task | P3  | helixtask: adversarial test, pure stateless decorator (no bug)                                                |
| HXC-1862 | Task | P3  | chutesaccount: adversarial test, read-only client no money-mutation sur                                       |
| HXC-1863 | Bug  | P2  | scheduler: Schedule/ScheduleOptimistic rank over map iteration -> non<br>placement on score ties              |
| HXC-1864 | Bug  | P2  |                                                                                                               |

| HXC      | Type | Pri | Title                                                                                                           |
|----------|------|-----|-----------------------------------------------------------------------------------------------------------------|
|          |      |     | infra: containerOrchestrator.Scale no negative-replicas guard -> slice-bo<br>(process crash)                    |
| HXC-1865 | Bug  | P2  | local: GPU.validate() lets NaN/Inf slip ordered guards -> poisons TCO,<br>forever                               |
| HXC-1866 | Task | P3  | phase7matrix: adversarial test, recompute + boundary correct, fail-closed                                       |
| HXC-1867 | Bug  | P2  | context: Detach leaks parent Deadline() despite 'never cancelled' -> self<br>open                               |
| HXC-1868 | Task | P3  | benchmark: adversarial test, CoefficientOfVariation Bessel/zero-mean co                                         |
| HXC-1869 | Task | P3  | compliancedoc: adversarial test, deterministic model card, no fabricated                                        |
| HXC-1870 | Task | P3  | e2eebench: adversarial test, nearest-rank percentile + ML-KEM key-agr<br>bug)                                   |
| HXC-1871 | Task | P3  | resources: adversarial test, CLAUDE-2 darwin parity VERIFIED REAL<br>completes pkg/ 196/196                     |
| HXC-1872 | Bug  | P1  | internal/node: UpdateNodeStatus mutates shared registry *Node pointer<br>race (live in gRPC server)             |
| HXC-1873 | Task | P3  | internal/gpu: adversarial test, GPU allocator no over-commit + CLAUD<br>(no bug)                                |
| HXC-1874 | Task | P3  | internal/gateway: adversarial test, authorize fail-CLOSED (no bug)                                              |
| HXC-1875 | Task | P3  | internal/security: adversarial test, identity/authz/policy all fail-CLOSED                                      |
| HXC-1876 | Bug  | P1  | internal/policy: scheduling rego permits non-numeric node.health -> ren<br>(schedules unhealthy node)           |
| HXC-1877 | Bug  | P2  | internal/messaging: Queue.Enqueue releases lock before send -> send-on<br>racing Close (teardown race)          |
| HXC-1878 | Task | P3  | internal/scheduler: adversarial test, gRPC ScheduleJob serialized no ove                                        |
| HXC-1879 | Task | P3  | internal/llm: adversarial test, Manager RWMutex-guarded + fail-closed                                           |
| HXC-1880 | Bug  | P2  | internal/build: SubmitBuild stores spec.BuildArgs by reference -> caller<br>worker read (shared-pointer alias)  |
| HXC-1881 | Bug  | P2  | internal/federation: scoreCell NaN score breaks cell-selection total order<br>admitted as Chosen                |
| HXC-1882 | Bug  | P1  | internal/health: aggregate switch has no default -> malformed/unknown<br>HEALTHY (fail-open readiness/liveness) |
| HXC-1883 | Task | P3  | internal/console: adversarial test, Registrar stateless no shared-pointer ra                                    |
| HXC-1884 | Task | P3  | internal/schema: adversarial test, structural verifier rejects malformed +<br>migrations (no bug)               |
| HXC-1885 | Task | P3  | internal/backup: adversarial test, round-trip byte-identical + correct argv<br>atomicity risks                  |
| HXC-1886 | Task | P3  | internal/trust: adversarial test, unknown node untrusted by default, NaN<br>bug)                                |
| HXC-1887 | Task | P3  | internal/advisory: adversarial test, distributed-lock service mutual exclus<br>(no bug)                         |
| HXC-1888 | Bug  | P2  | internal/costbroker: NaN price slips budget cap -> admitted, selected, po<br>spend to NaN (cap bypass)          |
| HXC-1889 | Task | P3  | internal/session: adversarial test, gRPC server stateless, Manager deep-c                                       |
| HXC-1890 | Task | P3  |                                                                                                                 |

| HXC      | Type    | Pri | Title                                                                                                                     |
|----------|---------|-----|---------------------------------------------------------------------------------------------------------------------------|
|          |         |     | internal/verifier: adversarial test, VerifyResult fail-CLOSED on zero/unlabeled tamper (no bug)                           |
| HXC-1891 | Task    | P3  | internal/chaos: adversarial test, single-use sequential runner, canary guard                                              |
| HXC-1892 | Bug     | P2  | cmd/htmux: streamSession 2 goroutines write same websocket.Conn w/ concurrent-write panic                                 |
| HXC-1893 | Task    | P3  | internal/wireguard: adversarial test, mesh RWMutex-guarded + dual key (no bug) — internal/ COMPLETE                       |
| HXC-1894 | Task    | P3  | cmd/helix-raftd: adversarial test, leader-gate 421 + validation defense-in-depth                                          |
| HXC-1895 | Task    | P3  | cmd/helix_infra: adversarial test, CLI arg parsers check err before value                                                 |
| HXC-1896 | Bug     | P2  | cmd/helix-session: runBridge 2 goroutines write same websocket.Conn w/ concurrent-write panic (server-side)               |
| HXC-1897 | Task    | P3  | cmd/helix-agent: adversarial test, validation fail-closed + teardown no-pause                                             |
| HXC-1898 | Task    | P3  | cmd/helix-policy: adversarial test, HTTP handler fail-CLOSED + HXC-1898 HTTP layer (no bug)                               |
| HXC-1899 | Task    | P3  | cmd/hxc-registry: adversarial test, enum validation + atomicity + numerical (no bug)                                      |
| HXC-1900 | Task    | P3  | cmd/helix-setup: adversarial test, config validation + YAML injection-safety (no bug)                                     |
| HXC-1901 | Bug     | P1  | cmd/helix-test: RunChallenge reports PASS+exit0 when evidence captured (CLAUDE-1 PASS-bluff in the QA harness)            |
| HXC-1902 | Task    | P3  | cmd/helix-health: adversarial test, /readyz /livez map non-Healthy -> 500                                                 |
| HXC-1903 | Task    | P3  | cmd/helix-raftctl: adversarial test, non-2xx/garbage response -> exit 1 (no bug)                                          |
| HXC-1904 | Task    | P3  | cmd/helixd: adversarial test, config fail-closed + clean shutdown + race-condition                                        |
| HXC-1905 | Bug     | P1  | WIRED hardenings: health fail-closed, metrics NaN-drop, swim saturation backup atomic write, session Reap                 |
| HXC-1906 | Bug     | P1  | ratelimit UserLimiter: NaN refill rate bypasses the rate-limit cap (fail-open)                                            |
| HXC-1907 | Task    | P2  | Adversarial coverage sweep: semaphore/lru/bursthysteresis/gpuattest/memleak/redundantexec (no prod bug; contracts pinned) |
| HXC-1908 | Bug     | P1  | build/cache DiskCache: path-traversal key + collision + empty-digest-tampering (integrity/security)                       |
| HXC-1909 | Task    | P2  | Adversarial coverage sweep waves 3-4: crdt/merkle, provider/{chutes,avro} backends (no prod bug; mutation-proven)         |
| HXC-902  | Task    | P1  |                                                                                                                           |
| HXC-903  | Task    | P1  |                                                                                                                           |
| HXC-904  | Feature | P0  | Phase 4 Build Service                                                                                                     |
| HXC-905  | Feature | P0  | SQLite HXC Registry                                                                                                       |
| HXC-908  | Bug     | P1  | Harden pkg/stonith IPMI credential handling (no -P on argv, redact in error)                                              |
| HXC-909  | Task    | P2  | pkg/multiraft: make RaftTransport async-delivery safe (Step under shared)                                                 |
| HXC-910  | Task    | P2  | pkg/deviceplugin: add concurrency test exercising Registry mutex under                                                    |
| HXC-911  | Bug     | P2  | pkg/chutes StreamChannel: honor cancellation on blocked reads + close                                                     |
| HXC-912  | Bug     | P2  | pkg/kraft: CreateTopic should return ErrTopicExists on conflicting re-cr                                                  |
| HXC-913  | Task    | P3  | pkg/porcupine: implement state-dedup memoization via Model.Equal fo                                                       |

| HXC     | Type     | Pri | Title                                                                                  |
|---------|----------|-----|----------------------------------------------------------------------------------------|
| HXC-915 | Task     | P2  | pkg/dst: wire 1000-sims/commit deterministic-simulation CI gate (GH A harness)         |
| HXC-916 | Task     | P2  | pkg/provider/chutes: ctx-interruptible backoff + honor Retry-After header              |
| HXC-917 | Bug      | P2  | pkg/multiraft: handle ShardStorage SetHardState/Append errors before                   |
| HXC-918 | Bug      | P2  | pkg/marketplaceadapter Akash: default MinReputation via constructor +                  |
| HXC-919 | Bug      | P2  | pkg/provider/runpod: surface Worker.Endpoint + don't hold lock across network call     |
| HXC-920 | Task     | P3  | pkg/provider/chutes: test parseRetryAfter HTTP-date branch (RFC1123 + past-date clamp) |
| HXC-927 | Task     | P2  | Surface multiraft durability persist-error through public API                          |
| HXC-934 | Bug      | P0  | Make cmd/e2ee-proxy consume security/pkg/e2ee instead of inline crypto                 |
| HXC-937 | Task     | P1  | Add e2e + chaos test types for consensus/federation/marketplace                        |
| HXC-938 | Task     | P2  | Add fuzz to crypto and benchmarks to hot paths                                         |
| HXC-939 | Docs     | P2  | Refresh stale per-phase GAP_AUDIT.md files (CLAUDE-3)                                  |
| HXC-940 | Task     | P2  | Wire the orphaned gate packages (covgate/archlint/etcdlint/qualitygate/p               |
| HXC-941 | Task     | P2  | Reconcile E2EE AEAD spec deviation (AES-256-GCM vs roadmap Cha                         |
| HXC-942 | Bug      | P1  | Add e2e/chaos -race integration tests for the 8 fixed concurrency hazard               |
| HXC-943 | Bug      | P1  | Test HelixQA validators (done) and vision/ORB (blocked) + fix helixqa                  |
| HXC-944 | Bug      | P0  | Fix 3 CRITICAL CVEs from trivy security scan                                           |
| HXC-945 | Bug      | P1  | Triage and resolve gosec HIGH findings (99 HIGH) from security scan                    |
| HXC-946 | Task     | P2  | Triage trivy misconfig HIGH (50) + suppress QA-sentinel secret false-p                 |
| HXC-947 | Task     | P3  | Add root go.mod require/replace for digital.vasic.security (out-of-works               |
| HXC-948 | Bug      | P2  | Fix api/v1 module path /v1-suffix blocking out-of-workspace require/rep                |
| HXC-950 | Bug      | P1  | sidecarutil HealthProbe load-robustness (WaitDelay + transient retry)                  |
| HXC-951 | Research | P2  | TLA+ spec for SWIM membership/failure-detection (TLC-verified safer                    |
| HXC-952 | Feature  | P2  | Unified hardware-inventory engine (CPU+memory+GPU+NPU aggrega                          |
| HXC-953 | Feature  | P2  | Real Raft-backed leader-election service (pkg/raftleader adapting pkg/ra               |
| HXC-954 | Feature  | P1  | Edge gateway multi-transport integration (QUIC + MQTT + WebSocket                      |
| HXC-955 | Feature  | P2  | Distributed KV demo over Raft (cmd/raftkv-demo) — building blocks c                    |
| HXC-956 | Feature  | P2  | OpenTelemetry single-runtime tracing foundation (observability/tracing)                |
| HXC-957 | Research | P2  | TLA+ spec for WireGuard/Noise-IK handshake (TLC-verified safety)                       |
| HXC-958 | Research | P2  | TLA+ spec for OR-Set CRDT convergence (TLC-verified, add-wins)                         |
| HXC-959 | Research | P2  | TLA+ spec for Two-Phase Commit atomicity (TLC-verified)                                |
| HXC-960 | Research | P2  | TLA+ spec for Saga compensation atomicity (TLC-verified)                               |
| HXC-961 | Research | P2  | TLA+ spec for vector-clock causal delivery (TLC-verified)                              |
| HXC-962 | Research | P2  | TLA+ spec for Raft single-server membership change safety (TLC-verifi                  |
| HXC-963 | Research | P2  | TLA+ spec for leader-lease leaseholder reads / no-stale-read (TLC-verif                |
| HXC-964 | Research | P2  | TLA+ spec for phi-accrual failure-detector accuracy (TLC-verified)                     |
| HXC-965 | Feature  | P1  | End-to-end cluster node-agent: compose discovery+raft+capabilities+ga (clusternode)    |

| HXC     | Type     | Pri | Title                                                                                                      |
|---------|----------|-----|------------------------------------------------------------------------------------------------------------|
| HXC-966 | Research | P2  | TLA+ spec for Chandy-Lamport distributed snapshot (TLC-verified)                                           |
| HXC-967 | Research | P2  | TLA+ spec for Lamport distributed mutual exclusion (TLC-verified)                                          |
| HXC-968 | Research | P2  | TLA+ spec for Dynamo-style quorum consistency $R+W>N$ (TLC-verified)                                       |
| HXC-969 | Feature  | P1  | clusternode real-WebSocket transport: cross-node messaging over real TCP                                   |
| HXC-970 | Research | P2  | TLA+ spec for Hybrid Logical Clock (TLC-verified)                                                          |
| HXC-971 | Research | P2  | TLA+ spec for Raft ReadIndex linearizable reads (TLC-verified)                                             |
| HXC-972 | Research | P2  | TLA+ spec for STONITH fencing safety / no dual-active (TLC-verified)                                       |
| HXC-973 | Research | P2  | TLA+ spec for gossip/epidemic dissemination convergence (TLC-verified)                                     |
| HXC-974 | Feature  | P1  | Real TCP-networked Raft for pkg/raft (network transport + leak-safe cluster)                               |
| HXC-975 | Research | P2  | TLA+ spec for PN-Counter CRDT convergence (TLC-verified)                                                   |
| HXC-976 | Feature  | P1  | Real BoltDB persistence for pkg/raft (durable on-disk store + recovery)                                    |
| HXC-977 | Research | P2  | TLA+ spec for rendezvous (HRW) consistent hashing minimal disruption                                       |
| HXC-978 | Feature  | P1  | pkg/raft: combined persistent+networked node (real TCP transport + on-disk store + restart-recover-rejoin) |
| HXC-979 | Research | P1  | TLA+ snapshot-install / log-compaction safety spec (RaftSnapshot), TLC-verified exhaustive                 |
| HXC-980 | Bug      | P1  | registry CLI: update auto-moves current_location on status change + record (doc-sync truthfulness)         |
| HXC-981 | Feature  | P1  | pkg/wasm: combined-bounds sandbox E2E (fuel + memory cap + capabilities simultaneously)                    |
| HXC-982 | Feature  | P0  | helix-raftd: real multi-process persistent Raft daemon (TCP + BoltDB) with from-disk E2E                   |
| HXC-983 | Docs     | P1  | docs/consensus.md: consolidated Raft consensus architecture doc + READER docs_chain tracking               |
| HXC-984 | Feature  | P1  | pkg/raft: snapshot/log-compaction InstallSnapshot integration test (runtime pairing of HXC-979)            |
| HXC-985 | Feature  | P1  | clusternode durable consensus: NodeAgent on persistent+networked raft with from-disk E2E                   |
| HXC-986 | Research | P1  | TLA+ Raft PreVote spec — stale/restarted node cannot disrupt a stable log (TLC-verified)                   |
| HXC-987 | Feature  | P1  | helix-raftd Prometheus /metrics endpoint (live raft state, real scrape proven)                             |
| HXC-988 | Feature  | P1  | e2ee adversarial tamper/wrong-key crypto tests (AEAD tag, ML-KEM-768 rejection)                            |
| HXC-989 | Docs     | P2  | helix-raftd observability materials: Prometheus scrape config + Grafana dashboard observability.md         |
| HXC-990 | Feature  | P1  | pkg/raft PreVote runtime disruption test (runtime pairing of HXC-986)                                      |
| HXC-991 | Feature  | P2  | helix-raftctl CLI client for helix-raftd admin API (live-cluster proven)                                   |
| HXC-992 | Feature  | P1  | pkg/scheduler Omega optimistic-concurrency barrier-race test (no double-count)                             |
| HXC-993 | Task     | P2  | gofmt hygiene: format 125 unformatted committed Go files across pkg/                                       |
| HXC-994 | Bug      | P0  | FIX real SWIM false-positive failure detection (healthy members wrong)                                     |
| HXC-995 | Feature  | P1  | pkg/antientropy end-to-end Merkle reconciliation test (only-differences-allowed)                           |
| HXC-996 | Feature  | P1  | pkg/mvcc snapshot-isolation + race-safety test (consistent point-in-time)                                  |

| HXC     | Type    | Pri | Title                                                                        |
|---------|---------|-----|------------------------------------------------------------------------------|
| HXC-997 | Feature | P1  | pkg/deltacrdt adversarial convergence tests + characterizes a real ORSet     |
| HXC-999 | Feature | P1  | pkg/splitbrain partition-safety test (at-most-one-authoritative across all c |